# Appendix C — Glossary of Key Terms

> Terms are defined at the level of precision required for production use. Abbreviations shown in parentheses. Cross-references in *italics*.

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## A

**AllGather** — A collective communication operation where each process contributes a shard, and after the operation every process holds all shards concatenated. Used in *ZeRO/FSDP* to reconstitute full parameter tensors before the forward pass. Opposite of *ReduceScatter*.

**AllReduce** — A collective communication operation that reduces (e.g., sums) tensors across all processes and gives every process the result. The backbone of *Data Parallelism* gradient synchronization. Implemented in *NCCL* using Ring or Tree algorithms, or via *NVLS/SHARP* for in-network aggregation.

**Arithmetic Intensity (AI)** — FLOPs performed per byte of memory traffic. AI = FLOPs / bytes. The key axis in the *Roofline* model. Operations with AI below the ridge point are *memory-bandwidth-bound*; above are *compute-bound*.

**Autoregressive Decoding** — The standard token generation algorithm: generate one token at a time, each conditioned on all prior tokens. Requires loading all model weights each step → inherently *memory-bandwidth-bound* at small batch sizes.

**AWQ (Activation-Aware Weight Quantization)** — A *PTQ* method that identifies "salient" weight channels (those corresponding to large-magnitude activations) and scales them before quantization to protect them from quantization error. Typically outperforms *GPTQ* for knowledge-intensive tasks. Uses per-group-128 granularity.

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## B

**BF16 (BFloat16)** — A 16-bit floating point format with 8 exponent bits and 7 mantissa bits. Same dynamic range as FP32 ($\pm 3.4 \times 10^{38}$) but lower precision. The standard training format for LLMs since GPT-3. Replaced *FP16* for training because its wider exponent range eliminates gradient overflow.

**Blocked Roofline** — See *Roofline Model*.

**Bubble Fraction** — The fraction of pipeline parallel compute time wasted on idle pipeline stages. Formula: `(PP-1) / (PP-1+M)` where PP = pipeline stages and M = microbatches. Target: < 10%.

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## C

**Chunked Prefill** — An inference optimization that splits long prompts into fixed-size chunks and interleaves prefill and decode steps within a single scheduler iteration. Reduces *TTFT* by sharing GPU time between prefill and decode rather than blocking decode requests during long prefill.

**Collective Operation** — A communication primitive that involves all processes in a group: *AllReduce*, *AllGather*, *ReduceScatter*, *Broadcast*, *All-to-All*.

**Compute-Bound** — A workload regime where the GPU's arithmetic units (Tensor Cores) are the bottleneck. Characterized by arithmetic intensity above the *roofline ridge point*. Optimization target: increase precision utilization (FP8), improve tile alignment, reduce kernel launch overhead.

**Continuous Batching** — An inference scheduling strategy that adds new requests to the active batch as soon as existing requests finish individual decode steps, rather than waiting for the entire batch to complete. Achieves near-100% GPU utilization without increasing TTFT. Standard in *vLLM*, *TRT-LLM*, *SGLang*.

**Context Parallelism (CP)** — A parallelism strategy that splits the sequence dimension across GPUs for long-context training. Each GPU holds *S/CP* query positions and attends over the full key-value set via All-to-All communication. Activates at S > 32K tokens.

**Copy-on-Write (CoW)** — A memory management technique in *PagedAttention* that allows multiple sequences to share the same physical KV cache blocks for their common prefix. Physical blocks are only copied when a sequence needs to write to a shared block (e.g., during *beam search* divergence).

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## D

**Data Parallelism (DP)** — A distributed training strategy where each worker holds a full model replica and processes different data shards. Gradients are synchronized across workers via *AllReduce* after each backward pass.

**DCGM (Data Center GPU Manager)** — NVIDIA's hardware telemetry daemon. Reads GPU silicon counters (SM utilization, temperature, ECC errors, power, NVLink bandwidth) at 1-second resolution. Exposes metrics to *Prometheus* via `dcgm-exporter`.

**Decode** — The autoregressive token generation phase of inference. Memory-bandwidth-bound at small batch sizes because the full model must be loaded from *HBM* for each generated token. Distinguished from *Prefill*.

**DPO (Direct Preference Optimization)** — A post-training alignment method that trains the policy model using a pair of chosen/rejected responses, without requiring a separate reward model. Simpler infrastructure than *PPO*: requires only policy + reference model (2× model memory vs PPO's 3-4×).

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## E

**ECC (Error Correction Code)** — A hardware mechanism in GPU *HBM* that detects and corrects memory bit errors. Single-bit errors (SBE / correctable) are fixed silently. Double-bit errors (DBE / uncorrectable) cause a *XID* error and indicate a failing GPU that must be replaced.

**Expert Parallelism (EP)** — A parallelism strategy for *MoE* models that distributes different experts across GPUs. Requires *All-to-All* communication to route tokens to the correct expert GPU. Typically deployed within the *NVLink* domain (EP ⩽ 8) to minimize latency.

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## F

**Fair-Share Scheduling** — A SLURM scheduling policy where job priority decreases as a user's historical resource consumption exceeds their allocated share. The fair-share score approaches 1.0 for under-utilizing users (high priority) and 0.0 for over-utilizing users (low priority). Prevents any team from monopolizing the cluster.

**FLOP (Floating Point Operation)** — One multiply or add operation. A multiply-accumulate (MAC) counts as 2 FLOPs. Used to measure compute requirements of neural network forward passes. Total training FLOPs ≈ 6 × N_params × N_tokens.

**FlashAttention** — A fused attention kernel (Dao et al. 2022) that implements attention in tiles, keeping intermediate K,V tiles in on-chip SRAM rather than materializing the full $O(S^2)$ attention score matrix in HBM. Reduces attention HBM traffic from $O(S^2)$ to $O(S)$. Enables training and inference at long context lengths.

**FP8** — An 8-bit floating point format. Two variants: E4M3 (4 exponent, 3 mantissa, range ±448, used for weights/activations) and E5M2 (5 exponent, 2 mantissa, range ±57K, used for gradients). Supported natively on H100/H200/B200 via *Transformer Engine*. Delivers ~2× throughput vs BF16 for GEMM.

**FSDP (Fully Sharded Data Parallel)** — PyTorch's native implementation of *ZeRO Stage 3*. Shards model parameters, gradients, and optimizer state across all DP workers. Equivalent to `ShardingStrategy.FULL_SHARD`. Requires 3× AllReduce bandwidth vs naive DP but enables much larger models per GPU.

**Fusion / Operator Fusion** — Merging multiple GPU operations into a single kernel to eliminate intermediate *HBM* round-trips. Vertical fusion: chain of sequential ops (RMSNorm + GEMM). Horizontal fusion: parallel ops (Q+K+V projections → QKV GEMM). Key optimization in *TorchInductor*, *TensorRT*, *NVFuser*.

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## G

**GEMM (General Matrix Multiplication)** — The core compute primitive of neural network layers. W × X for a linear layer. The dominant operation in *prefill* and training. Compute-bound at large batch sizes; memory-bandwidth-bound at small batch sizes (*decode*).

**GPUDirect RDMA** — An NVIDIA technology that allows the NIC to read from and write to GPU *HBM* directly via PCIe P2P, bypassing CPU stage buffers. Reduces communication latency by ~2 μs and frees ~12 CPU cores per DGX node that would otherwise serve as data proxy threads.

**GQA (Grouped Query Attention)** — An attention variant where multiple query heads share a single key-value head. Reduces *KV cache* size by H_q/H_kv (e.g., 8× for GQA-8 with 64 query heads). Standard in LLaMA-3, Mistral, Qwen. Enables much higher concurrent serving capacity vs *MHA*.

**GPTQ** — A *PTQ* method that applies second-order (Hessian-based) error correction during weight quantization. Processes weight matrices column by column, propagating quantization error to remaining columns to minimize output error. Achieves -1.3% MMLU delta for INT4 quantization of LLaMA-3 70B.

**Gradient Clipping** — Rescaling the gradient vector when its L2 norm exceeds a threshold (typically 1.0). Prevents gradient explosion during training. `torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)`.

**GRES (Generic Resources)** — SLURM's mechanism for tracking non-CPU/memory resources like GPUs. Specified with `--gres=gpu:h100:8` to request 8 H100 GPUs.

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## H

**HBM (High-Bandwidth Memory)** — 3D-stacked DRAM attached directly to the GPU die via a wide bus (1024-2048 bit). Provides much higher bandwidth than GDDR at the cost of lower capacity and higher cost per GB. Current standard: HBM3e at 4.8-8.0 TB/s.

**HFU (Hardware FLOP Utilization)** — Like *MFU* but accounts for recomputed activations. `HFU = achieved_FLOPs / (peak_FLOPs × time)` where achieved includes both forward FLOPs and recompute FLOPs. HFU ⩾ MFU always.

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## I

**Inference** — Running a trained model to generate outputs. Divided into *Prefill* (process the input prompt, KV-cache it) and *Decode* (generate output tokens autoregressively).

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## K

**KV Cache (Key-Value Cache)** — The cached tensor of Key and Value projections for all prior tokens in an autoregressive sequence. Stored in *HBM*. Enables $O(N)$ instead of $O(N^2)$ decode compute. Critical resource: KV cache size determines concurrent serving capacity.

**KV Cache Cascade** — A failure mode where high *KV cache* utilization (> ~88%) causes each new request to evict an existing sequence, which triggers eviction of another, exponentially growing the overhead. P99 *TTFT* can spike 10-50× in seconds. Prevention: admission control at 82% utilization.

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## L

**LoRA (Low-Rank Adaptation)** — A parameter-efficient fine-tuning method that adds trainable low-rank matrices ($A \in \mathbb{R}^{d\times r}$, $B \in \mathbb{R}^{r\times d}$) alongside frozen base model weights. Only A and B are updated during fine-tuning. Memory: proportional to $r \times d$ per layer, not $d \times d$.

**Lustre** — A widely-deployed parallel distributed filesystem for HPC and AI training clusters. Key optimization: `lfs setstripe -c -1 /path/` to stripe files across all OSTs (Object Storage Targets) for maximum write bandwidth. Must be set before data is written.

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## M

**MFU (Model FLOP Utilization)** — The fraction of theoretical GPU FLOPS actually used by the model. `MFU = (6 × N_params × tokens_per_second) / (N_GPUs × peak_FLOPS)`. The primary efficiency metric for training. Target: 45-60% on H100 clusters. Below 40%: significant bottleneck present.

**MHA (Multi-Head Attention)** — Standard attention with $H_q = H_{kv}$ query/key/value heads. Maximum quality, maximum KV cache. Largely replaced by *GQA* in modern LLMs for serving efficiency.

**MIG (Multi-Instance GPU)** — NVIDIA H100+ feature that partitions a single GPU into up to 7 isolated instances, each with dedicated SM slices and HBM bandwidth. Enables safe multi-tenant GPU sharing for inference of smaller models.

**MLA (Multi-Head Latent Attention)** — DeepSeek's architecture innovation that caches a compressed low-rank latent vector (d_c = 512) instead of full K,V heads. Reconstructs K,V at attention time from the latent. Achieves ~57× KV memory reduction vs standard MHA with near-zero quality loss. Enables 128K+ context serving.

**MoE (Mixture of Experts)** — An architecture that replaces each dense FFN layer with N expert FFNs, routing each token to K active experts (K << N). Compute ∝ active parameters; quality ∝ total parameters. DeepSeek-V3: 671B total, 37B active per token.

**MQA (Multi-Query Attention)** — Extreme version of *GQA* with H_kv = 1: all query heads share a single K,V. Maximum KV compression (~64× vs *MHA*) but noticeable quality degradation. Largely superseded by *GQA*.

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## N

**NCCL (NVIDIA Collective Communications Library)** — NVIDIA's library for GPU collective operations. Implements *AllReduce*, *AllGather*, *ReduceScatter*, *All-to-All*. Automatically selects Ring, Tree, or NVLS algorithms based on topology and message size.

**NF4 (NormalFloat 4-bit)** — A 4-bit quantization format (Dettmers et al.) optimized for normally-distributed weights. Uses a non-uniform quantization grid that places more quantization levels near zero, matching the approximate normal distribution of transformer weights. Used in *QLoRA*.

**NVFP4** — NVIDIA's 4-bit floating-point format for Blackwell (B200). 1-bit sign + 2-bit exponent + 1-bit mantissa with per-group (16 elements) FP8 scale factor. Peak: ~18 PFLOPS on B200. Enables ~8× throughput vs BF16 for weight-only inference.

**NVLink** — NVIDIA's high-bandwidth GPU-to-GPU interconnect implemented via NVSwitch chips on DGX nodes. Provides non-blocking all-to-all connectivity within an 8-GPU node. Bandwidth: 900 GB/s (NVLink 4.0, H100), 1.8 TB/s (NVLink 5.0, B200).

**NVLS (NVLink SHARP)** — In-network aggregation using compute units inside NVSwitch chips. Performs *AllReduce* summation in hardware, freeing GPU SMs from aggregation work. ~2× faster than Ring AllReduce for large messages. Auto-enabled on H100+ via NCCL ⩾ 2.16.

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## O

**Occupancy** — The ratio of active warps to maximum possible warps per SM. High occupancy allows the warp scheduler to hide memory latency by switching to ready warps. Low occupancy (caused by too many registers per thread or large SMEM allocation) can limit kernel performance.

**OST (Object Storage Target)** — An individual storage server in a *Lustre* filesystem. Each OST provides a fraction of the total bandwidth. Stripe count (`-c N`) determines how many OSTs a file is distributed across; use `-c -1` to stripe across all OSTs for maximum bandwidth.

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## P

**PagedAttention** — A *KV cache* memory management system (vLLM, 2023) that stores KV in fixed-size non-contiguous physical blocks with a per-sequence page table. Eliminates internal fragmentation (reduces waste from ~80% to ~1%) and enables *Copy-on-Write* prefix sharing. Industry standard in vLLM, TRT-LLM, SGLang.

**PFC (Priority Flow Control)** — IEEE 802.1Qbb Ethernet feature that enables lossless delivery for specific traffic classes. Sends PAUSE frames upstream to stop senders when a switch buffer fills. Required for *RoCEv2* to provide RDMA semantics (which require lossless delivery).

**Pipeline Parallelism (PP)** — A distributed training strategy that assigns different layers to different GPUs and pipelines microbatches through the stages. Enables very large models by splitting the depth of the network. Communication: point-to-point activation tensors between adjacent stages. See *Bubble Fraction*.

**PPO (Proximal Policy Optimization)** — The RL algorithm most commonly used for *RLHF*. Requires 4 concurrent systems: rollout workers (inference), reward scorer, reference model (frozen SFT), and policy updater. More infrastructure complexity than *DPO* but enables complex reasoning tasks.

**Prefill** — The phase of *inference* that processes the input prompt. Runs as a large batch GEMM on all prompt tokens simultaneously. Compute-bound at large batch/sequence sizes. Duration determines *TTFT*.

**Prefix Caching** — An inference optimization that stores KV cache blocks for common prefixes (system prompts, RAG documents) and reuses them across requests without recomputation. Reduces TTFT by 60-89% and KV memory by up to 91% for chatbot workloads with fixed system prompts.

**PTQ (Post-Training Quantization)** — Quantization applied after training is complete, using only a calibration dataset (128-512 samples). No retraining required. *GPTQ* and *AWQ* are PTQ methods.

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## Q

**QAT (Quantization-Aware Training)** — Quantization that simulates quantization error during training via "fake quantization" operators, allowing the model to adapt its weights to minimize

quality loss. Produces higher quality than *PTQ* at the cost of significant training compute (~10-20% of pretraining cost).

**QLoRA** — A parameter-efficient fine-tuning method combining *NF4* quantized base model with *LoRA* adapters. Enables fine-tuning 70B+ models on a single 48 GB GPU. Quality: ~93-94% of full fine-tune. Infrastructure: bitsandbytes library + PEFT.

**QoS (Quality of Service)** — *SLURM* scheduling tier configuration that sets resource limits, priority levels, and preemption rules per user class. Used to guarantee production workloads while allowing research jobs to fill remaining capacity.

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## R

**RadixAttention** — *SGLang*'s tree-structured prefix caching system. Stores KV cache blocks in a radix tree (trie), enabling physical block sharing for any shared prefix, not just fixed system prompts. Shared blocks have a single physical copy with reference counting. Eviction policy: leaves first (most recent shared prefixes retained longest).

**RDMA (Remote Direct Memory Access)** — A network transfer mechanism that allows a NIC to read from or write to remote GPU or CPU memory without involving the remote CPU. Provides low latency and low CPU overhead. Used in *InfiniBand* and *RoCEv2*.

**ReduceScatter** — A collective that reduces (sums) tensors and scatters the result: each process ends up with a disjoint shard of the total. Used in *ZeRO*/*FSDP* for gradient synchronization. Lower bandwidth cost than *AllReduce* for large models (each rank only receives its own shard).

**Ridge Point** — The *Arithmetic Intensity* at which the *Roofline* peak compute line meets the peak bandwidth line: `ridge = peak_FLOPS / peak_BW`. H100: 989T / 3.35T ≈ 295 FLOP/byte. Operations with AI < ridge are bandwidth-bound; AI > ridge are compute-bound.

**RLHF (Reinforcement Learning from Human Feedback)** — The process of fine-tuning a pretrained LLM using human preference data to align outputs with human values. Standard pipeline: SFT → Reward Model → PPO (or DPO/GRPO as simpler alternatives).

**RMSNorm** — A normalization layer that divides activations by their root-mean-square, without computing mean (unlike LayerNorm). Faster (one HBM pass vs two for LayerNorm) and equally effective. Standard in LLaMA-3, Mistral, Qwen. Key fusion opportunity with adjacent linear layers.

**RoCEv2 (RDMA over Converged Ethernet)** — RDMA over standard Ethernet fabric. Lower cost than *InfiniBand* but requires *PFC* for lossless delivery and active congestion control (*DCQCN* or *CTS*) to prevent incast. Used by Meta for Llama-3 training.

**Roofline Model** — A performance model that characterizes kernel performance by plotting achievable FLOPs/s vs arithmetic intensity. The "roof" has two segments: bandwidth-bound (slope = HBM_BW) and compute-bound (flat ceiling = peak_FLOPS). Used to classify every operation as bandwidth- or compute-bound and identify the correct optimization.

**RoPE (Rotary Positional Embedding)** — A position encoding method that rotates query and key vectors by a complex rotation corresponding to their position. Enables relative position encoding that generalizes better to longer sequences than learned absolute positions. Standard in LLaMA-3, Mistral.

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## S

**Scaling Efficiency** — The ratio of multi-GPU throughput to ideal linear scaling: `efficiency(N) = throughput(N GPUs) / (N × throughput(1 GPU))`. Target: 85%+ at 64 GPUs, 74%+ at 1,024 GPUs on H100 clusters. Below 80% at 64 GPUs: almost always a configuration problem.

**SHARP (Scalable Hierarchical Aggregation and Reduction Protocol)** — *InfiniBand* switch in-network AllReduce. Aggregation compute units inside Quantum2 switches perform gradient summation as packets flow through the fabric. Reduces AllReduce traffic by ~50%, latency by ~30-40%. Available on IB only (not RoCE).

**SLURM** — Simple Linux Utility for Resource Management. The dominant job scheduler for HPC and large-scale AI training. Three daemons: `slurmctld` (controller), `slurmdbd` (accounting database), `slurmd` (per-node agent).

**Speculative Decoding** — An inference optimization that uses a cheap *draft model* to propose K candidate tokens, then verifies them all with one target model forward pass. Acceptance criterion: `min(1, p_target/p_draft)`. Mathematically exact: output distribution is identical to target model. Speedup: 2-3.5× for code/SQL workloads.

**SwiGLU** — A gated linear unit activation function: $\mathrm{SwiGLU}(x) = \mathrm{SiLU}(W_1 x) \odot (W_3 x)$. Used in LLaMA FFN layers. Two GEMMs ($W_1$, $W_3$) followed by elementwise multiply. $W_1$ and $W_3$ can be fused into a single wider GEMM. Standard in LLaMA-3, Mistral, Qwen.

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## T

**Tensor Core** — Dedicated hardware units on NVIDIA GPUs that perform 4×4 or 8×16 matrix multiply-accumulate in one clock cycle. H100 achieves 989 TFLOPS BF16 using Tensor Cores vs ~20 TFLOPS without. Require specific matrix dimension alignments (multiples of 16 for BF16) for peak utilization.

**Tensor Parallelism (TP)** — A parallelism strategy that splits individual weight matrices across GPUs (column-parallel for FFN gate/up, row-parallel for FFN down and output projections). Requires *AllReduce* per transformer layer. Must be confined within the *NVLink* domain (TP ⩽ 8) to avoid IB latency bottleneck.

**TF32 (TensorFloat 32)** — A 19-bit compute format (8 exponent + 10 mantissa, stored in 32 bits) used transparently by A100+ Tensor Cores when FP32 input is provided. Delivers ~16× FP32 throughput with ~1% accuracy loss. Enabled by default in PyTorch 1.7+ via `allow_tf32=True`.

**TorchInductor** — The backend compiler in PyTorch 2.x (`torch.compile`) that converts traced FX graphs into optimized Triton kernels. Performs operator fusion, layout optimization, and tile size selection. Targets NVIDIA (via Triton → PTX), AMD (via Triton → HSACO), and CPU.

**TPOT (Time Per Output Token)** — The time between consecutive tokens during the *decode* phase. Primary latency metric for users during streaming generation. Formula: `TPOT = model_bytes / HBM_bandwidth` at small batch. H100, 70B BF16: ~42ms/token.

**Transformer Engine (TE)** — NVIDIA's library implementing automatic FP8 precision management. Maintains per-tensor rolling activation maximums (`amax_history`), computes dynamic scales, performs FP8 GEMM on Tensor Cores, and descales outputs back to BF16. Enables FP8 training and inference with a single code change.

**Triton** — An open-source Python-level GPU programming language (OpenAI). Abstracts SMEM management, warp scheduling, and autotuning. Used by TorchInductor for code generation and by engineers for custom GPU kernels without full CUDA C++ expertise.

**TTFT (Time to First Token)** — The latency from request submission to generation of the first output token. Dominated by *prefill* time. The primary user-perceived latency for interactive LLM services. Target SLA typically P50 < 300ms, P99 < 800ms.

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## V

**vLLM** — An open-source high-performance LLM serving library (UC Berkeley, 2023). Pioneered *PagedAttention*. Features: continuous batching, prefix caching, speculative decoding, FP8 KV cache, disaggregated prefill-decode.

**Volcano** — A Kubernetes batch scheduling framework that adds gang scheduling (all pods start together or none start), queue management, and priority preemption for distributed ML training jobs. Prevents GPU deadlocks from partial rank allocation.

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## W

**Warp** — The CUDA execution unit: 32 threads that execute the same instruction simultaneously

on an SM. SIMT (Single Instruction, Multiple Thread) model. All 32 threads in a warp must execute the same instruction at each clock cycle (divergence = serialization).

**Wavefront** — AMD's equivalent of a CUDA warp, but with 64 threads (2× NVIDIA). Affects kernel tile sizing and reduction loop depth for AMD GPU programming.

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## X

**XID Error** — NVIDIA GPU error code logged to the kernel message buffer (`dmesg`). XID 48 (uncorrectable ECC) and XID 95 (uncontained error) require immediate GPU replacement. XID 74 (NVLink error) requires topology investigation.

**XLA (Accelerated Linear Algebra)** — Google's ahead-of-time compiler for JAX/TensorFlow models. Compiles from *HLO* (High-Level Operations) IR to PTX or TPU assembly. All shapes must be statically known at compile time. The only compiler supported on Google TPUs.

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## Z

**ZeRO (Zero Redundancy Optimizer)** — DeepSpeed's memory sharding framework with three stages. Stage 1: shard optimizer state. Stage 2: + shard gradients. Stage 3: + shard model parameters. Stage 3 enables models of arbitrary size across any number of GPUs at the cost of 3× AllReduce bandwidth vs naive DP. Implemented in PyTorch as *FSDP*.

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## Quick Acronym Index

```
AI      Arithmetic Intensity          MHA    Multi-Head Attention
AWQ     Activation-aware Weight Quant MIG    Multi-Instance GPU
BF16    BrainFloat16                  MLA    Multi-Head Latent Attention
CP      Context Parallelism           MoE    Mixture of Experts
DCGM    Data Center GPU Manager       MQA    Multi-Query Attention
DCE     Dead Code Elimination         MFU    Model FLOP Utilization
DCP     Distributed CheckPoint        NCCL   NVIDIA Collective Comms Library
DP      Data Parallelism              NF4    NormalFloat 4-bit
DPO     Direct Preference Optimization NVLS  NVLink SHARP
ECC     Error Correction Code         NVFP4  NVIDIA FP4 (Blackwell)
EP      Expert Parallelism            OST    Object Storage Target
FP8     8-bit Floating Point          PFC    Priority Flow Control
FSDP    Fully Sharded Data Parallel   PP     Pipeline Parallelism
GEMM    General Matrix Multiply       PPO    Proximal Policy Optimization
GQA     Grouped Query Attention       PTQ    Post-Training Quantization
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| | | | |
|---|---|---|---|
| GPTQ | GPTQ quantization method | QAT | Quantization-Aware Training |
| HBM | High-Bandwidth Memory | RDMA | Remote Direct Memory Access |
| HFU | Hardware FLOP Utilization | RLHF | RL from Human Feedback |
| IB | InfiniBand | RMSNorm | Root Mean Square Normalization |
| KV | Key-Value (cache) | RoPE | Rotary Positional Embedding |
| LoRA | Low-Rank Adaptation | SDC | Silent Data Corruption |
| MFU | Model FLOP Utilization | SLA | Service Level Agreement |
| MBR | Model Batch Resource | SM | Streaming Multiprocessor |
| SMEM | Shared Memory | SWA | Sliding Window Attention |
| TE | Transformer Engine | TP | Tensor Parallelism |
| TF32 | TensorFloat 32 | TPOT | Time Per Output Token |
| TRT | TensorRT | TTFT | Time to First Token |
| XID | NVIDIA GPU Error Code | ZeRO | Zero Redundancy Optimizer |

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*— End of Appendix C —*

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